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Original Article

Retrieval-augmented generation enhances large language model performance on the Japanese orthopedic board examination

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ABSTRACT

Introduction: Large language models (LLMs) have shown potential in medical applications. However, their effectiveness in specialized medical domains remains underexplored. The integration of Retrieval-Augmented Generation (RAG) has been proposed to improve these models by reducing hallucinations and enhancing domain-specific information access. Through this evaluation, we aim to assess whether RAG can effectively bridge the gap between LLMs' current capabilities and the accuracy needed for medical use by examining GPT-3.5 Turbo, GPT-4o, and o1-preview on the 2024 Japanese Orthopedic Specialist Examination.

Methods: A specialized database was created using the "Standard Textbook of Orthopedics", and GPT-3.5 Turbo, GPT-4o, and o1-preview were evaluated with and without RAG. Models were tested on text-based and image-based questions exactly as presented in Japanese. An error analysis was conducted to identify key performance factors.

Results: GPT-3.5 Turbo showed no substantial improvement with RAG, with its overall accuracy remaining at 28 %, compared to its baseline of 29 % without RAG. GPT-4o rose from 62 % to 72 %, while o1-preview increased from 67 % to 84 %. Error analysis indicated that GPT-3.5 Turbo primarily failed to apply retrieved data, whereas GPT-4o and o1-preview made errors when the database lacked relevant information or when dealing with image-based questions.

Conclusions: The integration of RAG significantly boosted performance for GPT-4o and especially o1-preview. While both models surpassed the passing threshold, o1-preview demonstrated a level of proficiency relevant to clinical practice. However, RAG did not improve performance on GPT-3.5 Turbo because it lacks effective reasoning abilities.

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1. Introduction

Large Language Models (LLMs), such as GPT-4, have demonstrated significant capabilities in complex tasks across various fields, including medicine [1–3]. As the potential for LLMs in clinical practice grows, medical examinations are increasingly being used as benchmarks to assess their applicability and performance in real-world settings [4,5].

These models have achieved passing scores on general medical exams such as the United States Medical Licensing Examination (USMLE), with accuracies of 60–64 % [6,7]. However, their performance declines when confronted with more specialized medical domains. For instance, GPT-4 has shown limitations in pharmacy and radiology, achieving lower accuracy rates on questions requiring deep domain knowledge [8,9]. Similarly, on standardized urology assessments, GPT-4 and GPT-3.5 scored 44 % and 30 % respectively, particularly struggling with domain-specific questions [10]. These trends indicate that while LLMs perform reasonably well on general medical knowledge assessments, their effectiveness diminishes as

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the level of specialization increases, highlighting challenges in their clinical application.

The accuracy limitations of LLMs in specialized domains can be attributed to either insufficient domain-specific knowledge or impaired retrieval of existing knowledge. This limitation results in 'hallucination' [11,12], which is the generation of convincing but incorrect information. Retrieval-Augmented Generation (RAG) technology has emerged as a potential solution to these fundamental challenges [13,14]. RAG couples the generative capabilities of an LLM with a retrieval system that queries a verified knowledge base during response generation. By incorporating specific, trusted sources, RAG supplements the model's inherent knowledge with domain-specific information, reducing hallucinations and improving accuracy [15,16]. Despite these advances, there remains a notable gap in our understanding of RAG technology's effectiveness in improving LLM accuracy within specialized medical domains where precision and reliability are essential. Furthermore, while RAG enhances LLM performance, the inherent capabilities of the underlying LLM remain important. Therefore, evaluating advanced models like o1-preview in conjunction with RAG is important to understand the potential of this technology in specialized fields.

This study aims to investigate whether RAG can enhance LLM performance in a complex, highly specialized medical context—the Japanese Orthopedic Specialist Examination. We compare three prominent LLMs—GPT-3.5 Turbo, GPT-4o, and o1-preview—on the 2024 version of this exam, which is administered fully in Japanese and demands in-depth orthopedic knowledge. We also perform an error analysis of the models' responses to identify key failure patterns and guide potential strategies for future improvements. Through this approach, we aim to benchmark their performance in clinically relevant tasks, using domain-specific medical exams as settings that simulate clinical practice.

2. Materials and methods

2.1. Study context and examination

This study evaluated the performance of three LLMs—GPT-3.5 Turbo, GPT-4o, and o1-preview—in answering questions from the 2024 Japanese Orthopedic Specialist Examination. The examination, administered in January 2024, is a comprehensive assessment for orthopedic surgery board certification in Japan. It consists of multiple-choice questions where candidates must select one, two, or three correct options out of five. GPT-3.5 Turbo has a knowledge cutoff of September 2021, while GPT-4o and o1-preview both have a cutoff of October 2023, ensuring that the 2024 exam content effectively tests their ability to address recently published materials.

2.2. Data preparation

A specialized database was created using text data from the "Standard Textbook of Orthopedics, 15th Edition" (referred to as "Hyojun Seikeigeka" in Japanese), a widely used reference that aligns closely with the content of the Japanese Orthopedic Specialist Examination. The text data was uploaded to a private, access-restricted GitHub repository, ensuring that only authorized prompts could retrieve the necessary information. The content was segmented into chunks of 1000 tokens with an overlap of 50 tokens to ensure continuity and context retention across chunks. Data preparation and retrieval were conducted on Google Colaboratory, operating in a closed environment to maintain the integrity of the database.

2.3. Model and RAG setup

We used three versions of OpenAI's language models: GPT-3.5-turbo (gpt-3.5-turbo-0125), GPT-4o (gpt-4o-2024-05-13), and o1-preview (o1-preview-2024-09-12). The RAG system was implemented using the text-embedding-ada-003 model for generating embeddings. In this context, an embedding is a way of converting text into numerical data that a machine learning model can understand. This conversion allows the model to recognize patterns and relationships in the text, making it easier to retrieve relevant information. The implementation was carried out on Google Colaboratory, ensuring that the model's responses were derived solely from the database content, with no access to external information.

2.4. Question input and classification

The exam prompts were input into the models exactly as they appeared in the Japanese Orthopedic Specialist Examination. No modifications were made to the prompts, ensuring that the models were tested under conditions as close to the actual exam as possible, including retaining the original Japanese language of the prompts. We then classified the questions according to whether they contained images (image-based vs. text-based) and by the type of inquiry (knowledge-based vs. case-based). Finally, we noted whether each item required selecting a single correct choice or multiple correct choices. Table 1 provides an overview of these classifications.

2.5. Evaluation

Answer correctness was defined strictly: a response was considered correct only if the model selected all and only the correct options. Responses with any missing or extra choices were marked as incorrect. Accuracy rates were directly compared across models.

2.6. Error analysis

Error analysis was conducted on the RAG-augmented models, categorizing errors into five types: (1) Missing Database Information: Required information was not present in the database. (2) Retriever Issues: The system failed to retrieve relevant chunks from the database. (3) Failed to Utilize Retrieved Information: RAG retrieved relevant information, but the model did not effectively use it to answer correctly. (4) Image Interpretation Errors: Incorrect medical interpretation of images was made. (5) Case-Based Question Errors: Incorrect diagnosis was given for case-based questions.

2.7. Statistical analysis

We first performed Cochran's Q test to examine whether there were any overall differences among the six model configurations (GPT-3.5 Turbo, GPT-3.5 Turbo + RAG, GPT-4o, and GPT-4o + RAG,

Table 1

Classification of questions by image presence, case type, and answer format.

Category	Subcategory	Count
Image/Text	Image-based questions	38
	Text-based questions	62
Case/Knowledge	Case-based questions	34
	Knowledge-based questions	66
Multiple/Single	Multiple-choice	52
	Single answer	48

Table 2
Accuracy for o1-preview, GPT-4o and GPT3.5 Turbo with and without RAG.

	o1-preview	o1-preview + RAG	GPT-4o	GPT-4o + RAG	GPT-3.5	GPT-3.5 + RAG
Overall Accuracy (%)	67	84	62	72	29	28
Image-based questions (%)	53	74	47	61	34	18
Text-based questions (%)	76	90	69	79	26	34
Case-based questions (%)	56	76	50	56	38	21
Knowledge-based questions (%)	73	88	68	80	24	32
Multiple-choice (%)	75	87	60	63	29	23
Single Choice (%)	58	81	65	81	29	33

GPT, Generative Pre-trained Transformer; RAG, Retrieval-Augmented Generation.

o1-preview, o1-preview + RAG). If the Cochran's Q test indicated a statistically significant difference ($p < 0.05$), we then conducted pairwise McNemar tests to identify which specific model pairs showed significant performance discrepancies. We considered a p -value < 0.05 as indicative of statistical significance.

2.8. Ethics

This study did not involve any human subjects or animal experiments. The research utilized publicly available orthopedic board examination questions and artificial intelligence models. No patient data or personal information was used in this study.

3. Results

For GPT-3.5 Turbo, without RAG, the model achieved an accuracy of 34 % on image-based questions and 26 % on text-based questions, resulting in an overall accuracy of 29 %. When RAG

was applied, image-based accuracy decreased to 18 % while text-based accuracy increased to 34 %, yielding an overall accuracy of 28 %.

For GPT-4o, without RAG, the model achieved an accuracy of 47 % on image-based questions and 69 % on text-based questions, with an overall accuracy of 62 %. When RAG was applied, the accuracy improved to 61 % for image-based questions and 79 % for text-based questions, resulting in an overall accuracy of 72 %.

For the o1-preview model, an overall accuracy of 67 % was achieved without RAG, with 53 % accuracy on image-based questions and 76 % on text-based questions. When RAG was enabled, performance increased to an overall accuracy of 84 % (74 % on image-based and 90 % on text-based questions). Detailed results for each model are provided in Table 2.

Fig. 1 displays the response provided by o1-preview + RAG for Question 1, summarizing the model's reasoning and final answer selection.

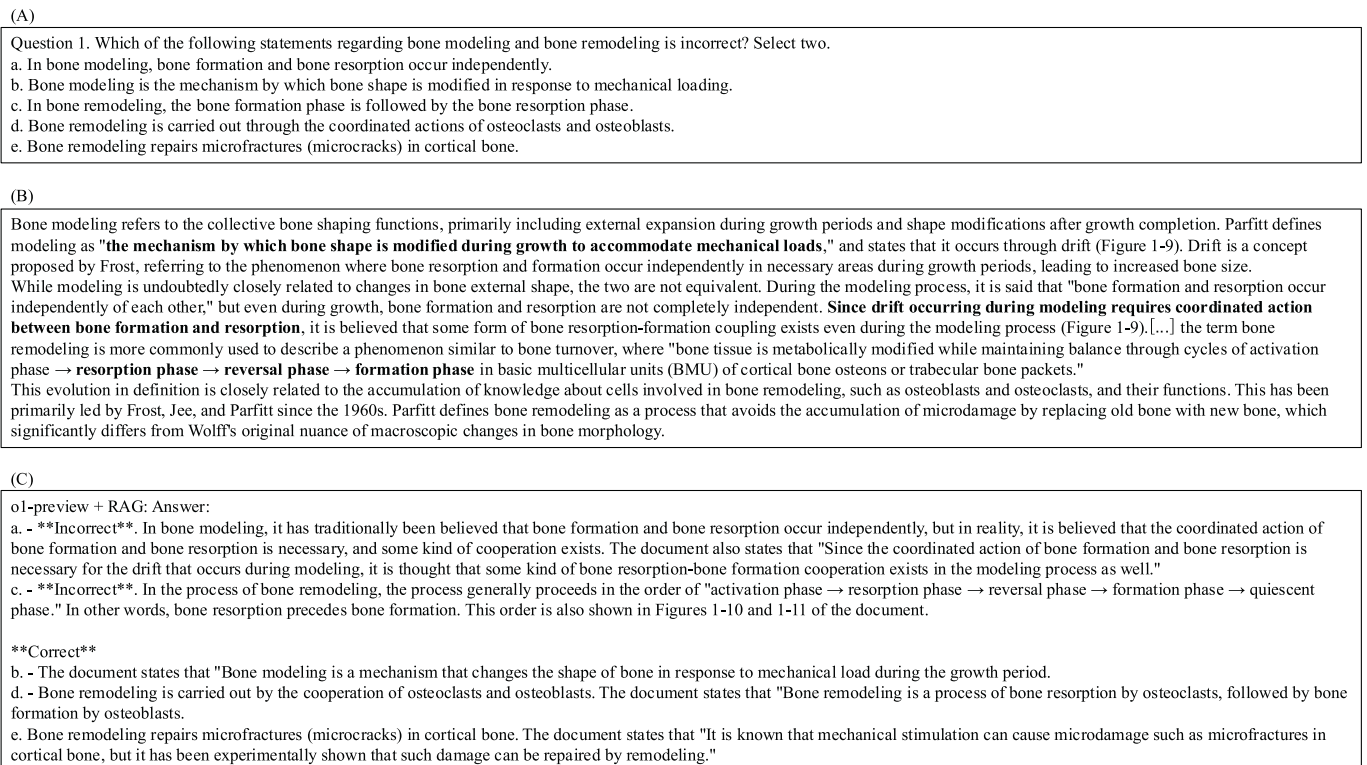


Fig. 1. (A) Question from the Japanese Orthopedic Specialist Examination, a non-image, knowledge-based, multiple-selection question (translated from Japanese). (B) Retrieved chunk containing relevant information from external sources used by the model to generate a response (translated from Japanese; note: the ellipsis ([...]) indicates that part of the retrieved text has been omitted for brevity). This example represents one of the 10 chunks retrieved by RAG. The original output has been manually modified to highlight key points with bold lines in figure. (C) Model response generated by o1-preview + RAG, indicating which statement is incorrect (translated from Japanese). RAG, Retrieval-Augmented Generation.

We used Cochran's Q test to examine whether the six model configurations differed overall. The result was statistically significant ($Q = 135.652$, $p < 0.000001$). Consequently, we performed pairwise McNemar tests to compare each pair of models. Representative pairwise comparisons indicated no significant differences between GPT-4o and GPT-4o + RAG (corrected $p = 0.662570$) and between GPT-4o and o1-preview (corrected $p = 1.000000$). In contrast, a significant difference was found between o1-preview and o1-preview + RAG (corrected $p = 0.012738$).

Error analysis for GPT-3.5 Turbo + RAG revealed 19 errors requiring image interpretation, 25 stemming from a failure to utilize retrieved information, and 14 resulting from missing data in the database. In contrast, GPT-4o + RAG incurred 12 text-based errors due to missing database information, 3 caused by retriever issues, and 12 related to image interpretation. Lastly, o1-preview + RAG exhibited 11 errors that required detailed image interpretation and 5 that arose from absent knowledge in the database (Table 3).

4. Discussion

This study reveals the benefits of combining RAG with LLMs for specialized medical exams. Augmented with RAG, the model's overall accuracy rose considerably, particularly for o1-preview, which improved from 67 % to 84 %. GPT-4o also exhibited a clear gain from 62 % to 72 %, highlighting the value of real-time domain knowledge. Conversely, GPT-3.5 Turbo showed little improvement with RAG, likely due to limitations in reasoning and Japanese processing, suggesting RAG efficacy depends on LLM capabilities.

When RAG was applied, the model's overall accuracy significantly improved, particularly in the case of o1-preview, where the accuracy increased from 67 % to 84 %. Previous studies have examined the performance of LLMs with their native pre-existing knowledge, with GPT-4 achieving an accuracy rate of 59–61 % on the Japanese Orthopedic Specialist Examination [17], and a similar performance level (61 %) on the Orthopaedic Surgery In-Training Examination (OITE), performing at a level comparable to a third-year orthopedic surgery resident [18]. While GPT-4 established these benchmarks, our study expands upon these findings by evaluating o1-preview. Notably, o1-preview not only exceeded these earlier GPT-4-level benchmarks with a baseline accuracy of 67 %, but also achieved an overall accuracy of 84 % (with text-based accuracy reaching 90 %) when combined with RAG. The application of RAG in our study demonstrated significant improvements over the performance of LLM alone, even with unmodified Japanese questions and without specific prompt engineering, where any unclear or ambiguous responses were marked as incorrect. The errors observed with GPT-4o + RAG, which also appeared in o1-preview + RAG at a lower frequency, were categorized into three main types. The first type involved cases where the required information was not present in the database. The second type involved retriever errors, where relevant chunks were not effectively retrieved, often due to the high frequency of specific keywords, such as "rheumatoid arthritis," which overwhelmed the

retrieval system. These issues are caused by the current model's token limitations and processing speed constraints. The third type of error was related to image-based questions. The inclusion of image-based questions posed significant challenges for the models. While GPT-4o and o1-preview demonstrated some ability to interpret images, its capacity to extract precise medical findings remained limited [17]. Despite the limitations in image interpretation, both GPT-4o and o1-preview were able to derive correct answers by analyzing the textual components of questions and multiple-choice options. For image-based questions, GPT-4o achieved an accuracy of 61 % and o1-preview achieved an accuracy of 74 %. These results suggest that the models can partially overcome their limited image interpretation capabilities through textual analysis. However, the overall improvement in accuracy for image-based questions remained limited, highlighting the challenges in effectively dealing with visual content.

Regarding GPT-3.5 Turbo, unlike GPT-4o and o1-preview which showed significant improvements with RAG, this model demonstrated limited improvement with RAG implementation due to its weaker reasoning capabilities. It was observed that the model frequently failed to apply knowledge correctly, even when the relevant chunks were retrieved. This suggests that the model's reasoning performance is limited, leading to errors in utilizing the retrieved information effectively [3]. The poor performance can be attributed to GPT-3.5 Turbo's lower accuracy in processing Japanese input compared to English input [19]. Moreover, some questions that were correctly answered by the model without RAG were incorrectly answered when RAG was applied, due to the model's tendency to rely heavily on the retrieved context, especially when interpreting complex or nuanced Japanese language content [20]. Complex contexts or nuanced language in the retrieved chunks appeared particularly challenging for GPT-3.5 Turbo to interpret accurately. This contrasts with the enhanced performance observed in GPT-4o and o1-preview, highlighting the critical role of the underlying LLM's capabilities in the effectiveness of RAG.

To further improve accuracy, several strategies could be employed. First, the organization of the database could be refined. Currently, chunks are uniformly divided into 1000-token segments. However, chunking can affect the effectiveness of RAG systems, and further refinement, such as adjusting chunk sizes or exploring content-based segmentation, may enhance retrieval accuracy [21,22]. Additionally, re-ranking retrieved chunks to place the most relevant ones first could improve response quality. Future work should consider these adjustments to optimize the balance between chunk size, re-ranking, and retrieval performance. Secondly, prompt engineering plays a crucial role in enhancing large language model performance, particularly for complex tasks, as demonstrated across various professional exams [2,23]. Previous studies have shown that optimizing prompt structure and standardizing output format improved model performance in the Japanese National Medical Licensing Examination [24]. This highlights the importance of careful prompt design in enhancing model performance, especially for complex tasks involving nuanced medical

Table 3

Error Analysis for o1-preview, GPT-4o and GPT3.5 Turbo with RAG.

Error Type	o1-preview + RAG	GPT-4o + RAG	GPT-3.5 + RAG
Information not available in the database	5	12	14
Retriever error (incorrect chunk retrieval)	0	3	7
Unable to utilize retrieved information despite RAG	0	1	25
Need for accurate image interpretation	11	12	19
Failure to diagnose in case-based questions	0	0	7

GPT, Generative Pre-trained Transformer; RAG, Retrieval-Augmented Generation.

knowledge. Furthermore, Med Prompt, developed by Microsoft researchers [25], also exemplifies the effectiveness of specialized prompt designs in medical question answering. Med Prompt utilizes a structured prompt format incorporating question decomposition and knowledge integration to improve reasoning. However, OpenAI's documentation indicates that simpler prompts are often optimal for models like o1-preview (<https://platform.openai.com/docs/guides/reasoning/advice-on-prompting>). Complex strategies like few-shot learning or chain-of-thought may not enhance, or even hinder, its performance. Therefore, exploring and adapting prompt engineering techniques, while considering the specific LLM and task, remains crucial for further performance enhancement. Regarding methodological approaches, fine-tuning represents one established approach [26]. This process involves further training a pre-trained LLM on specific task or domain-related data and adjusting the model's internal parameters to integrate new knowledge. However, it does not necessarily enhance the model's ability to learn or retain new information effectively [27]. Therefore, RAG was chosen for this study due to its superior ability to leverage up-to-date and accurate knowledge, particularly for specialized medical information.

This study has some limitations. The performance of the models was evaluated based on the 2024 Japanese Orthopedic Specialist Examination, which was the most recent available test at the time of this study. Relying on a single year of examination limits the generalizability of the results, as the performance may vary with different test versions or formats. The decision to use the 2024 examination was guided by the knowledge cutoffs of GPT-3.5 (September 2021), GPT-4o (October 2023), and o1-preview (October 2023). This approach ensured that the models were evaluated on content that they had not encountered during training, thereby reducing potential biases. Future studies should include multiple exam years, incorporate raw imaging data. Additionally, the study did not include image inputs for image-based questions, which may have limited the models' ability to interpret and respond accurately to such questions. These limitations suggest that further research is needed to validate the findings across a wider range of medical exams and to refine the RAG system to reduce any biases that might affect the models' performance.

Despite these limitations, this study demonstrates a significant advancement in applying LLMs to specialized medical domains. The successful integration of RAG technology with o1-preview, demonstrating substantial improvement in accuracy, offers promising potential for enhancing AI capabilities in complex, knowledge-intensive fields. Nevertheless, further improvements are needed to fully bridge the gap between LLMs' current capabilities and the accuracy needed for medical use.

5. Conclusion

In conclusion, the integration of RAG technology significantly improved the performance of both GPT-4o and o1-preview on a specialized medical exam, with both models surpassing the passing threshold—although the performance gain was more pronounced in o1-preview. These results indicate that RAG can enhance the proficiency of language models to a level that may be relevant to clinical practice. However, as observed with GPT-3.5 Turbo, the effectiveness of RAG is limited when the model's reasoning performance is insufficient. Future work should focus on optimizing the database structure, enhancing retrieval algorithms, improving image interpretation capabilities, and developing more effective prompt designs. Through these improvements, LLMs are expected to become more reliable and powerful tools in specialized domains, including medicine.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Anthropic's Claude and OpenAI's ChatGPT to assist with proofreading and grammatical checks of the English text. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no conflict of interest.

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